4. Multimodal Visual Understanding + Visual Line Following

Introduction

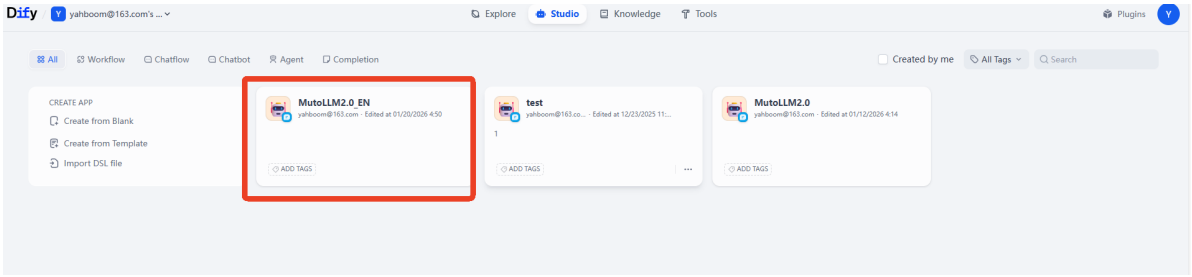
In this tutorial, we will learn how to give instructions by chatting with Muto in the Dify interface.

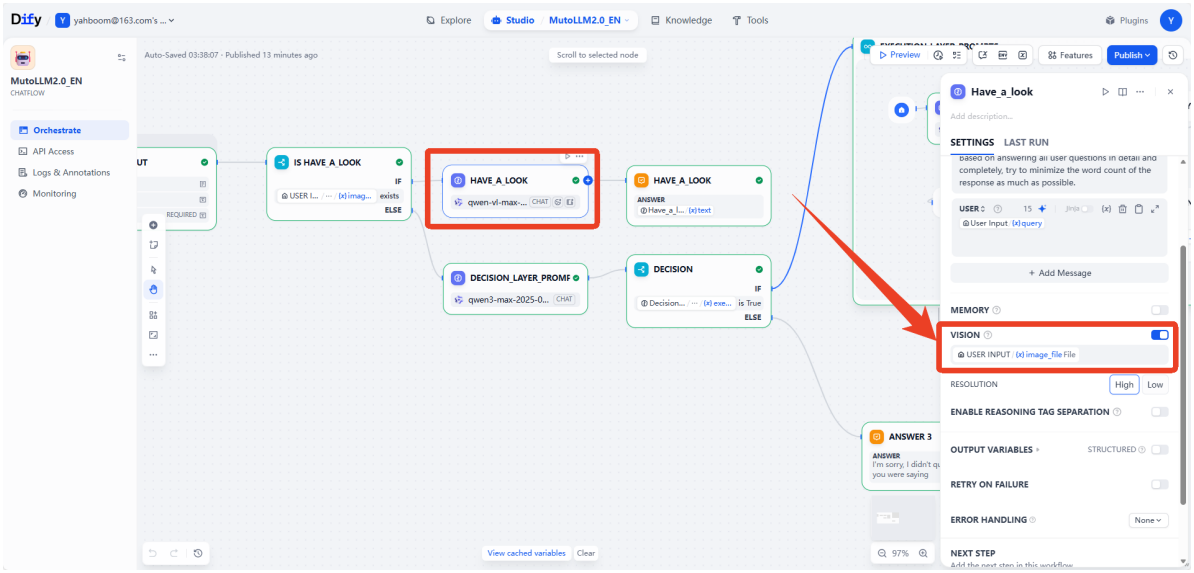
Steps

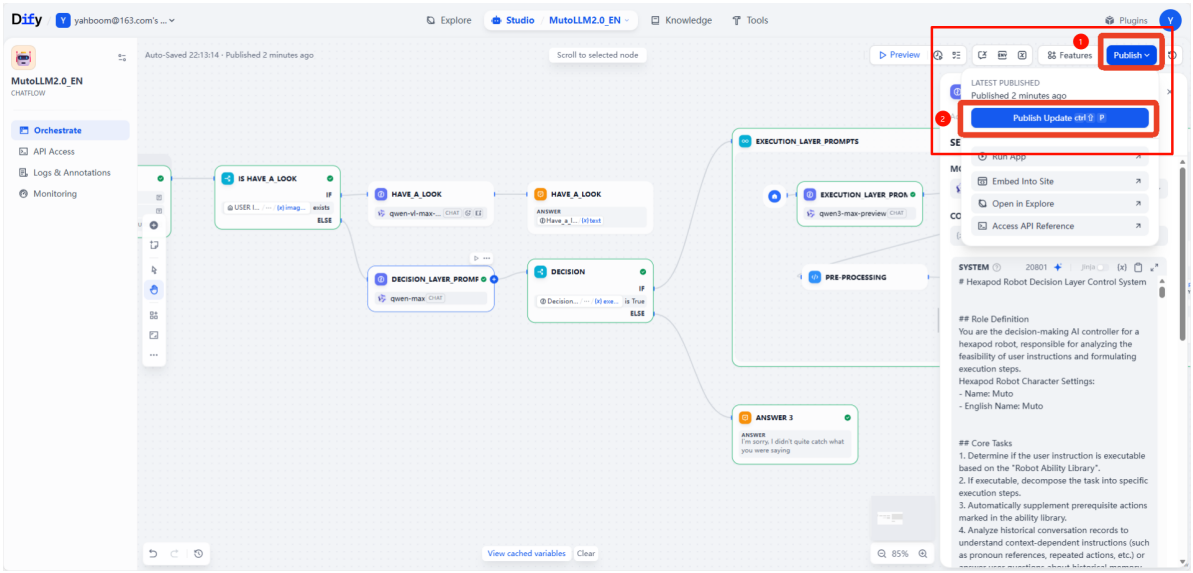
Prerequisites

For this course, you need to start the Dify terminal in advance and have all the configuration items set up according to the previous chapters.

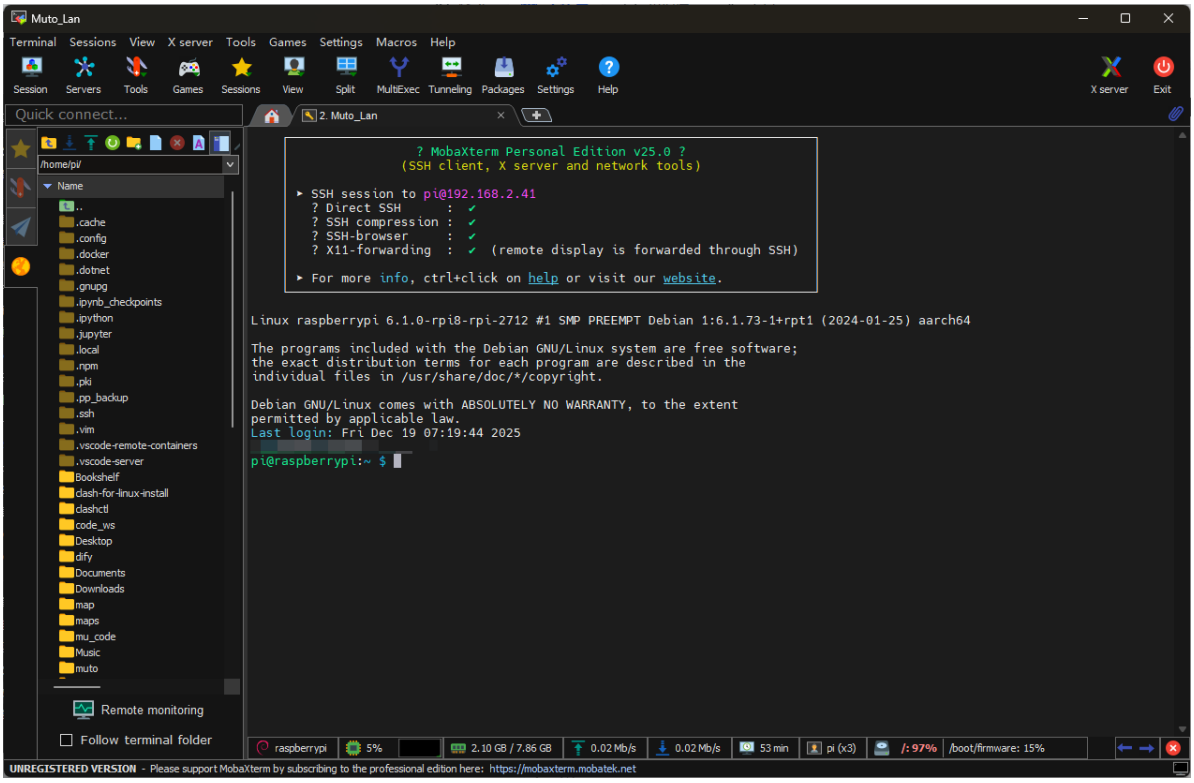
```
#!/usr/bin/env bash

# ==========================================
# User Configuration Area
# ==========================================

# 1. Voice Module Configuration
# ============================

# 1.1 Provider & Keys
export ALIBABA_API_KEY="sk-[redacted]20"

export FORCE_VOICE_PROVIDER="xunfei"

# 1.2 General Voice Settings
export VOICE_LANGUAGE="en"

# 2. Dify Configuration
# Auto-detect IP from wlan0
export DIFY_HOST=$(ip addr show wlan0 2>/dev/null | grep "inet " | awk '{print $2}' | cut -d/ -f1 | head -n 1)
# Fallback if wlan0 has no IP
if [ -z "$DIFY_HOST" ]; then
    export DIFY_HOST="127.0.0.1"
fi
```

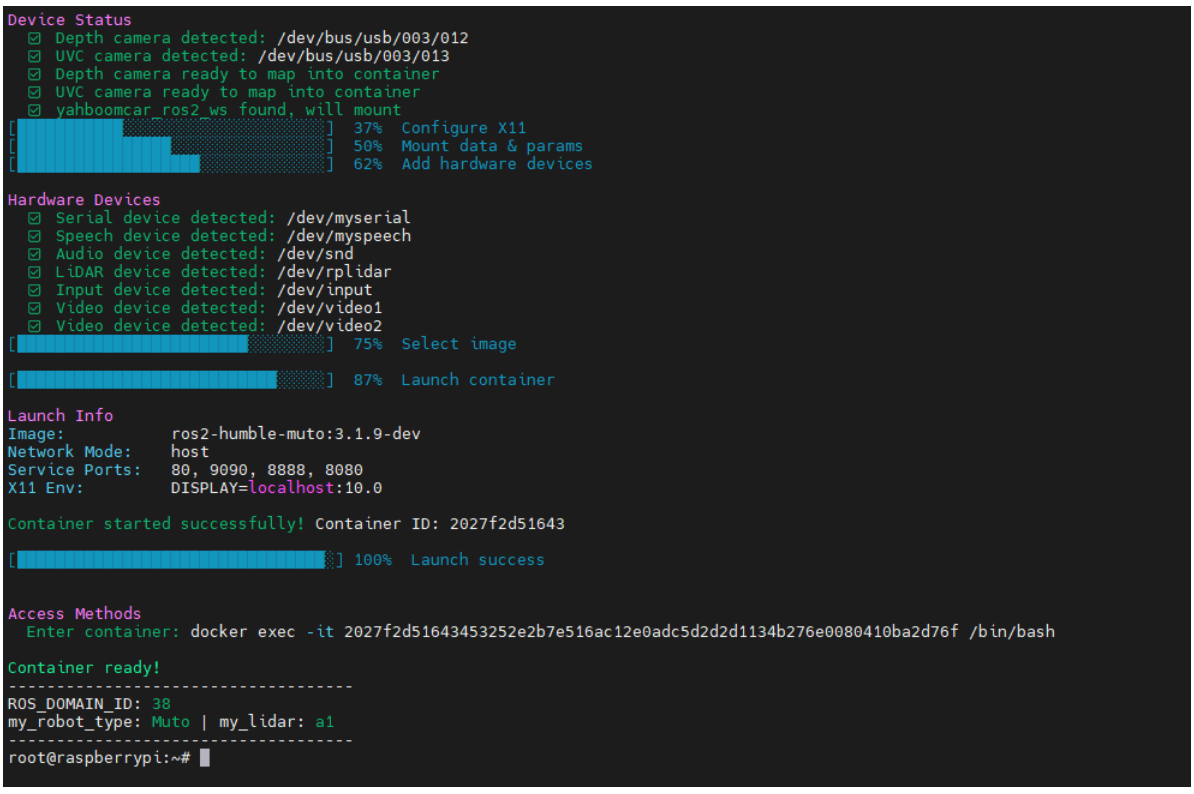






This course requires MutoRS to be powered on. Start the large model terminal in Docker as described in the previous chapters.





Use the command to start the large model backend.

```
./rebuild_and_launch.sh
```

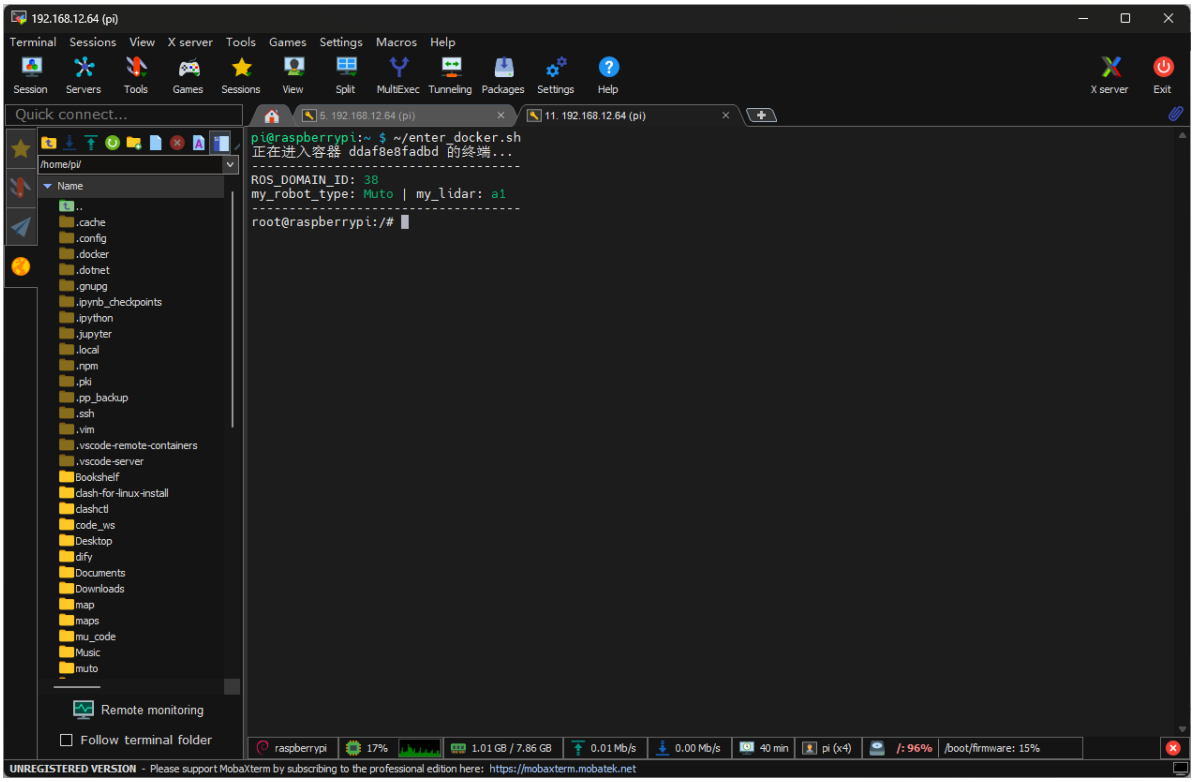
```
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.voice_interface - INFO - Speech to text initialized succes
sfully with alibaba provider
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.core.text_to_speech - INFO - TTS Provider alibaba
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor st
arted
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor st
arted
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - Audio queue manager initialized s
uccessfully
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - All voice modules initialized suc
cessfully
[muto_controller_node-1] [INFO] [1766137692.315363643] [muto_controller_node]: Command executor initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315920470] [muto_controller_node]: Attempting to prestart camera system...
[muto_controller_node-1] ✗ Node not running: /camera/camera
[muto_controller_node-1] Current running nodes: ['/muto_controller_node', '/navigation_manager']
[muto_controller_node-1] [INFO] [1766137697.078459054] [muto_controller_node]: Camera service started successfully: Camera
 started successfully
[muto_controller_node-1] [INFO] [1766137697.085229984] [persistent_image_capture_3188dc8d]: Persistent image capture node
initialized for topic: /camera/color/image_raw with Best Effort QoS
[muto_controller_node-1] [INFO] [1766137697.086642747] [muto_controller_node]: Persistent camera node initialized: Camera
node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.087635385] [muto_controller_node]: FastAPI application created successfully
[muto_controller_node-1] [INFO] [1766137697.149655757] [muto_controller_node]: API routes registered successfully
[muto_controller_node-1] 2025-12-19 09:48:17,149 - voice_module.core.voice_wake - INFO - Starting voice wake detector on p
ort: /dev/myspeech
[muto_controller_node-1] 2025-12-19 09:48:17,154 - utils.serial_comm - INFO - Serial data receiving started
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.core.voice_wake - INFO - Voice wake detector started succe
ssfully
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.voice_interface - INFO - Wake detection started
[muto_controller_node-1] [INFO] [1766137697.155236161] [muto_controller_node]: Voice assistant auto-start: True
[muto_controller_node-1] [INFO] [1766137697.156658443] [muto_controller_node]: ROS2 publisher created successfully
[muto_controller_node-1] [INFO] [1766137697.157379009] [muto_controller_node]: Status timer and health check created succe
ssfully
[muto_controller_node-1] [INFO] [1766137697.157921559] [muto_controller_node]: Muto Controller Node initialized successful
ly
[muto_controller_node-1] [INFO] [1766137697.158425850] [muto_controller_node]: Starting FastAPI server on 0.0.0.0:8080
[muto_controller_node-1] [INFO] [1766137697.161259210] [muto_controller_node]: FastAPI server started successfully
[muto_controller_node-1] INFO:     Started server process [221]
[muto_controller_node-1] INFO:     Waiting for application startup.
[muto_controller_node-1] INFO:     Application startup complete.
[muto_controller_node-1] INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
```

Getting Started

In this section, we will explain the visual line following feature.

The visual line following feature requires opening a new terminal in the background, entering Docker, and running the visual line following processing center.

```
enter_muto_docker
```



In the new terminal, execute:

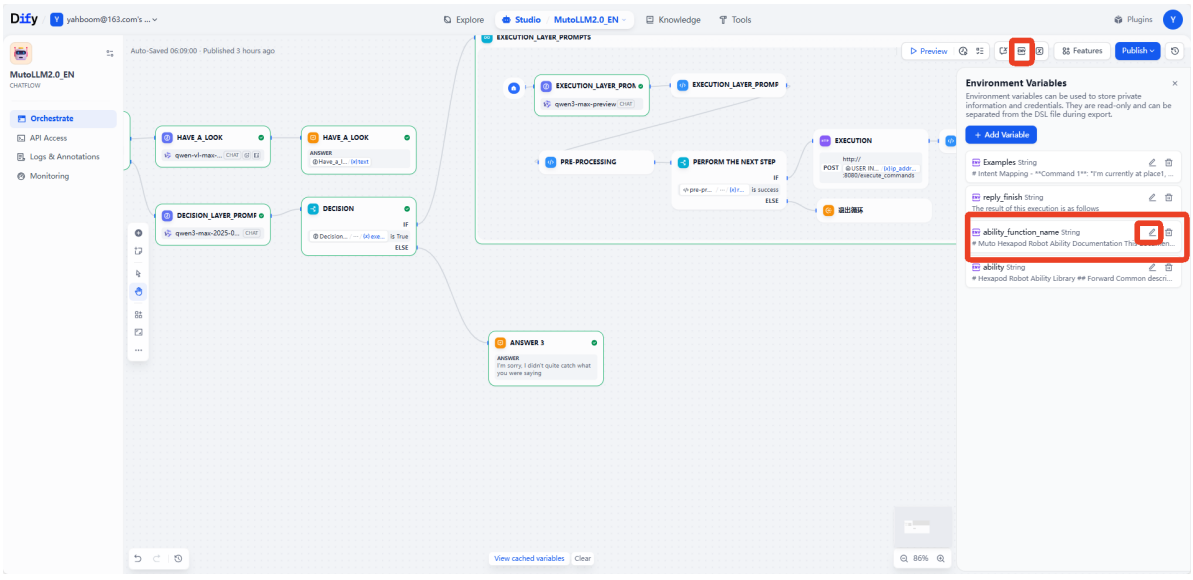
```
ros2 launch yahboomcar_linefollow follow_line_launch.py
```

You can see that a window like this has opened.

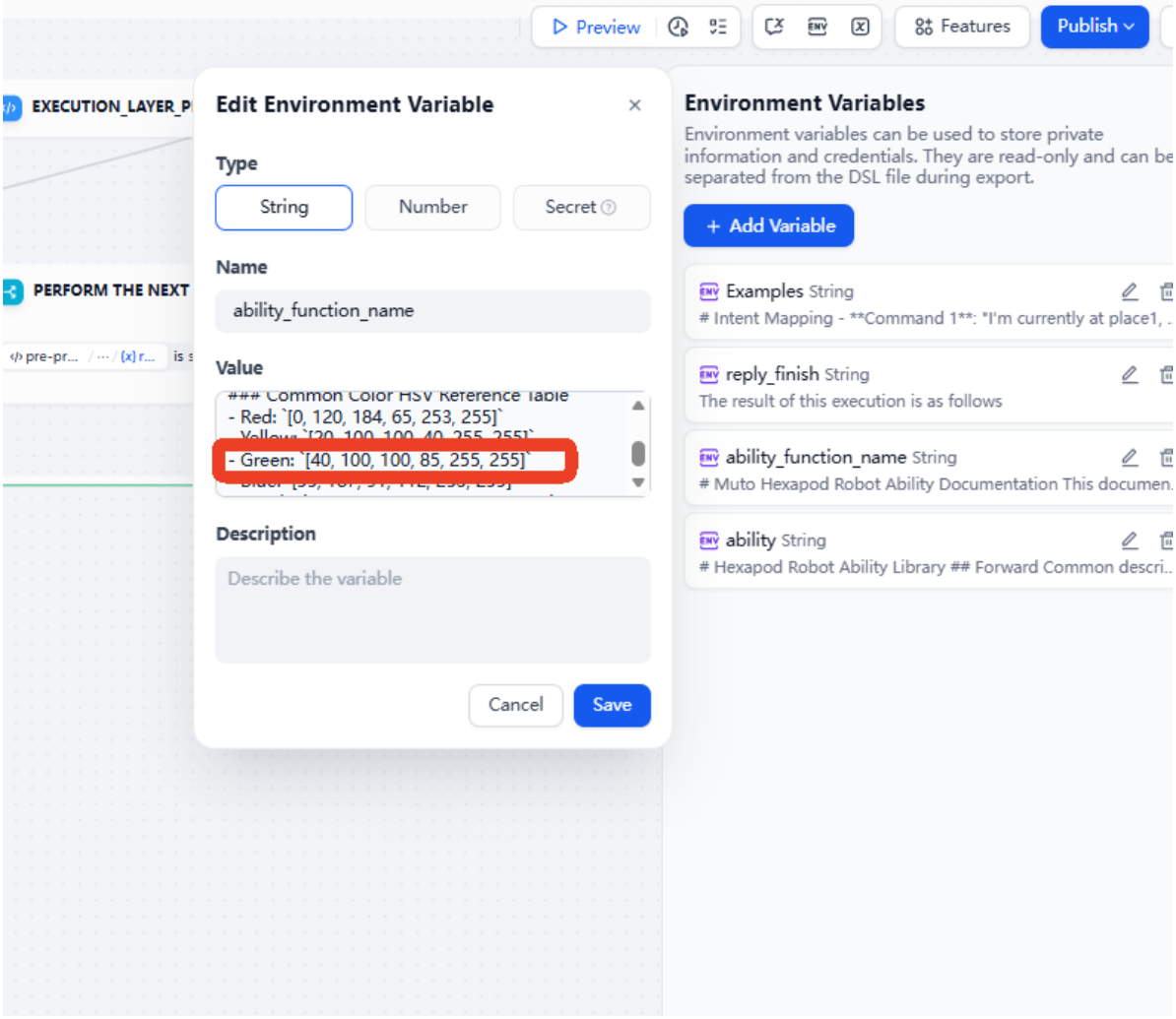
Use the left mouse button to select the yellow line to get its HSV range.

The HSV in the upper left corner is [35, 94, 96, 89, 253, 255].

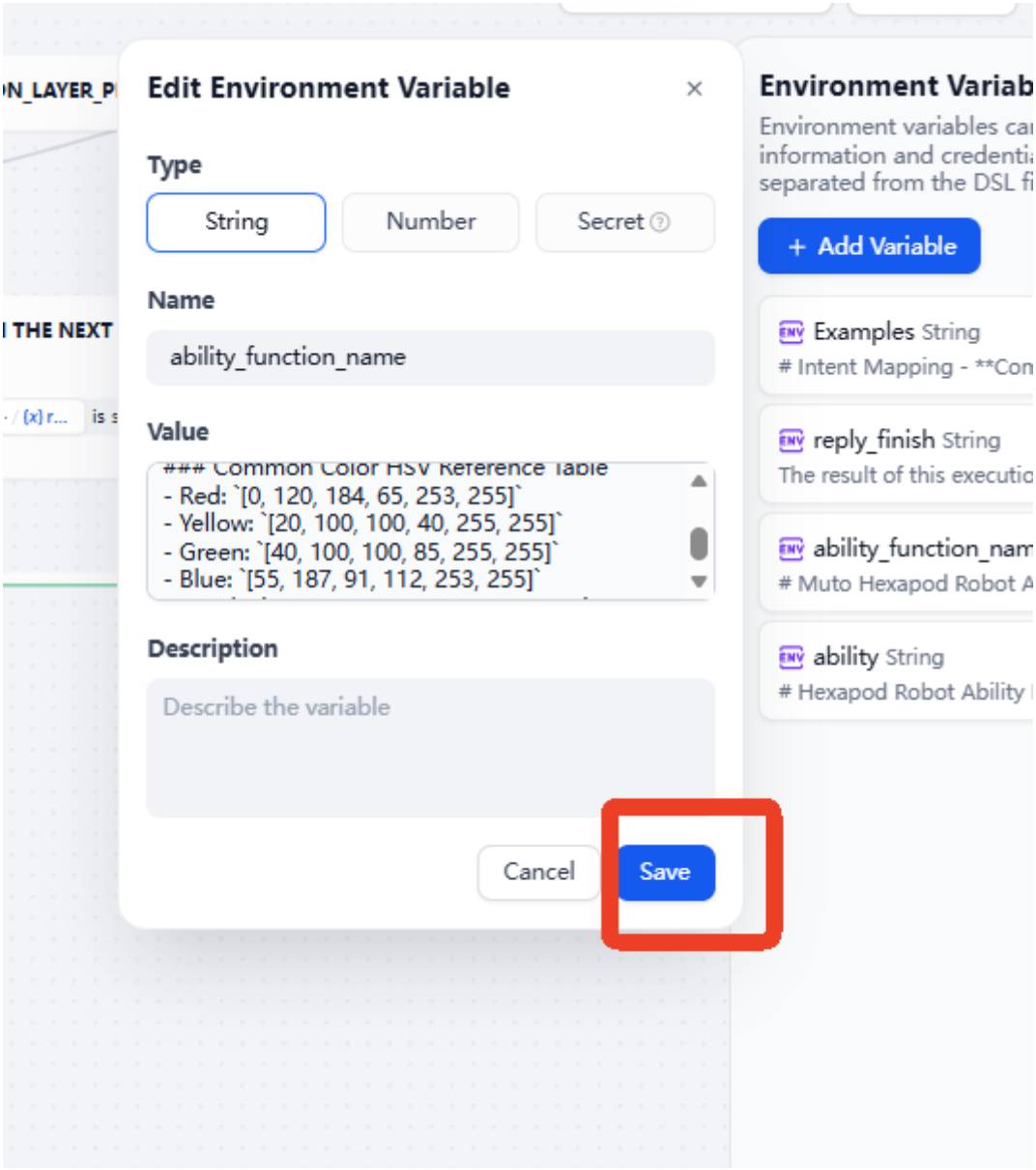
Copy this HSV range, open Dify, find the env icon in the upper right corner, and edit our `ability_function_name` variable.



Edit the HSV value here, changing the yellow HSV value to the one we just obtained.



Click save after modification.

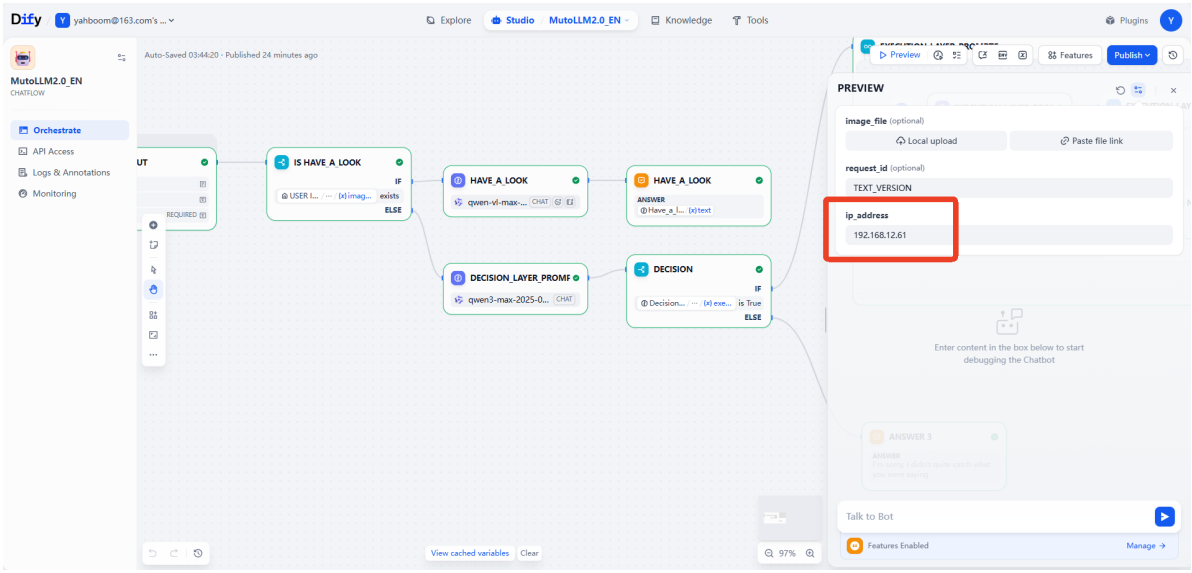


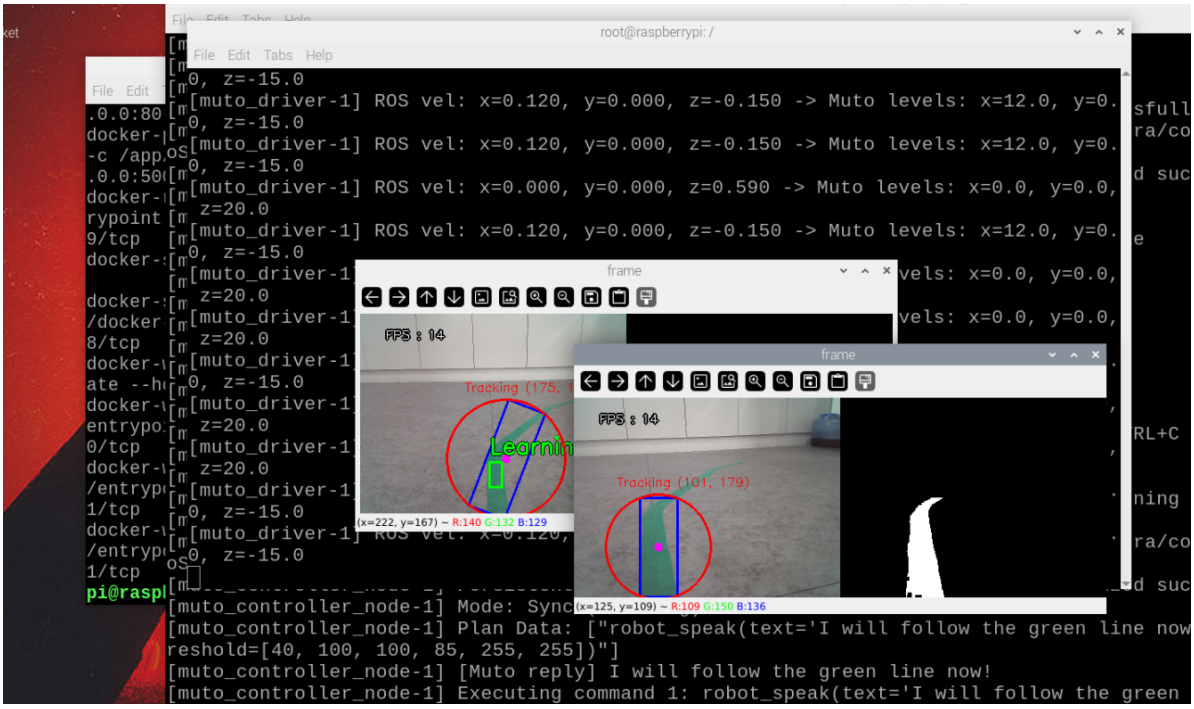
The command we use in this section is:

> "Drive along the xx line in front of you."

We can interact with Muto by voice (the illustration shows text interaction, the voice effect is the same).

You can see that Muto will announce the voice command and then start moving forward along the line automatically.


Dify
Explore
Studio
MutoLLM2.0_EN
Knowledge
Tools
Plugins
MutoLLM2.0_EN
CHATFLOW
Orchestrate
API Access
Logs & Annotations
Monitoring
Auto-Saved 03:44:20 · Published 24 minutes ago
Preview
Features
Publish
IS HAVE A LOOK
IF
USER I... imag... exists
ELSE
HAVE A LOOK
qwen-vl-max-...
CHAT
HAVE A LOOK
ANSWER
Have_a_l... text
DECISION_LAYER_PROMF
qwen3-max-2025-0...
CHAT
DECISION
IF
Decision... exe... is True
ELSE
PREVIEW
image_file (optional)
Local upload
Paste file link
request_id (optional)
TEXT_VERSION
ip_address
192.168.12.61
Enter content in the box below to start debugging the Chatbot
Talk to Bot
Features Enabled
Manage
View cached variables
Clear
97%


root@raspberrypi: /
File Edit Tabs Help
[muto_driver-1] ROS vel: x=0.120, y=0.000, z=-0.150 -> Muto levels: x=12.0, y=0.
[muto_driver-1] ROS vel: x=0.000, y=0.000, z=0.590 -> Muto levels: x=0.0, y=0.0,
z=20.0
frame
FPS : 14
Tracking (175,
Learnin
(x=222, y=167) ~ R:140 G:132 B:129
frame
FPS : 14
Tracking (101, 179)
(x=125, y=109) ~ R:109 G:150 B:136
[muto_controller_node-1] Mode: Sync
[muto_controller_node-1] Plan Data: ["robot_speak(text='I will follow the green line now
reshold=[40, 100, 100, 85, 255, 255])"]
[muto_controller_node-1] [Muto reply] I will follow the green line now!
[muto_controller_node-1] Executing command 1: robot_speak(text='I will follow the green